Computer Systems and Internet Technologies - COMP1589 Lab 1

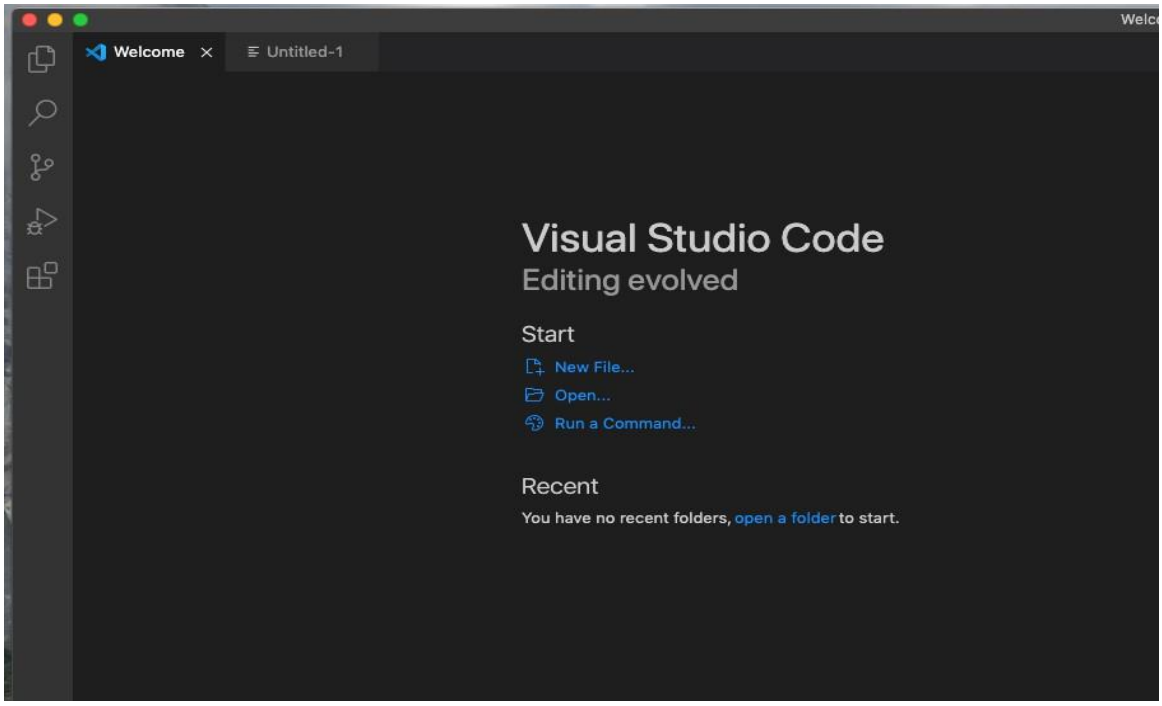
A HOME PAGE AND BASIC HTML

Using VS code to create a home page

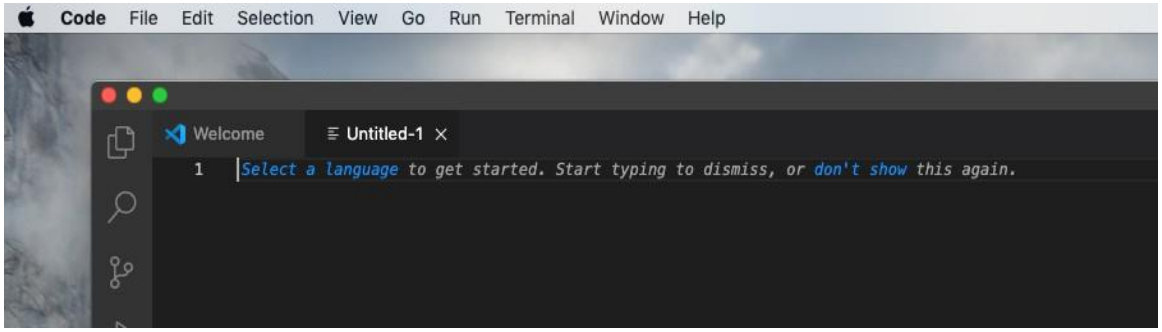
We will look at how to create a basic web page in VS code. This has lots of advantages over notepad but there are many text editors you can use to create HTML pages.

If you have not already done so, on your G drive or One Drive create folders for your 4 courses. Make sure there is one for COMP 1589 and create a sub folder week1.

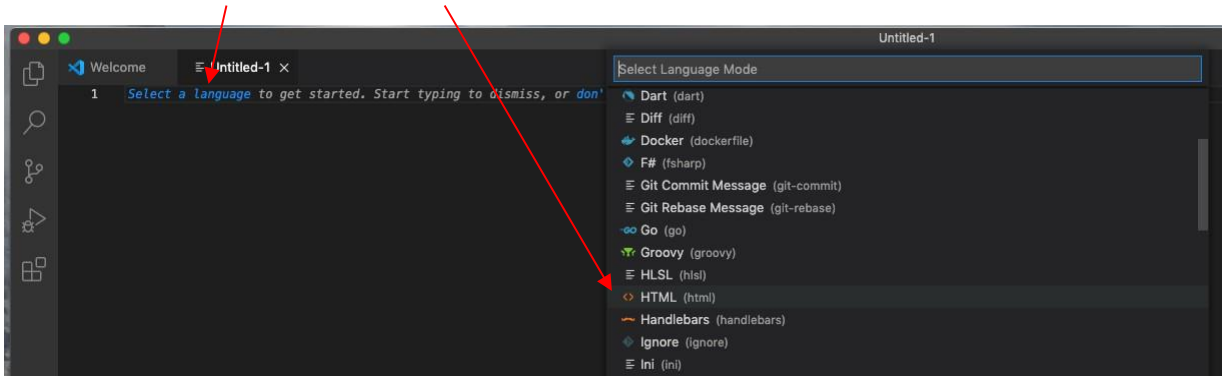
Open VS code



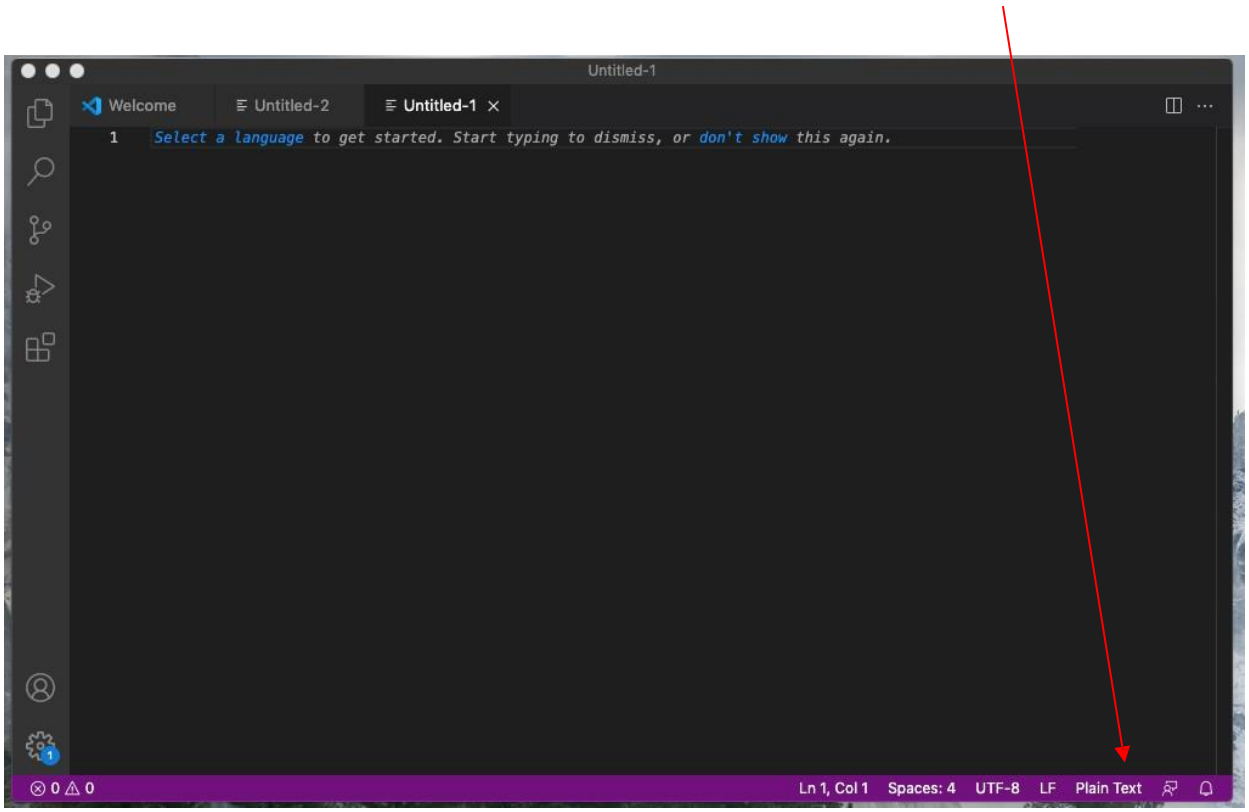
Goto > file new or double click in the tab bar at the top of the page and you should get this



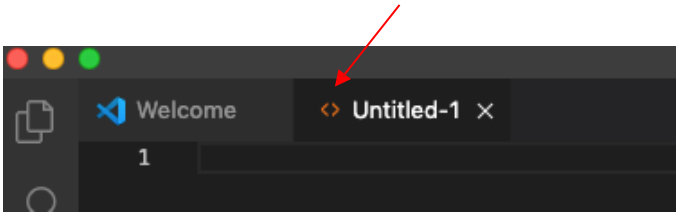
Where it says select a language select html from the list



If that option was not available if you look at the bottom right of the page it will say "plain text" you can also click here to bring up the select language menu.

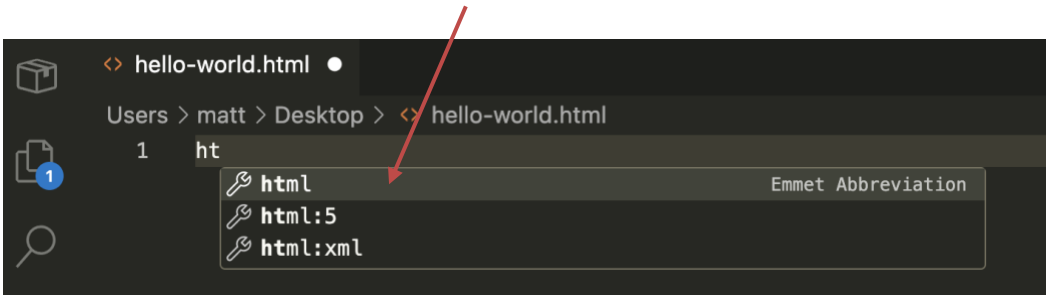


You can tell your new doc it now HTML because of the little orange tag icon



A home page

1. On the very first line of your new document begin to type html. A menu should pop up. We want the 2nd item **html:5**

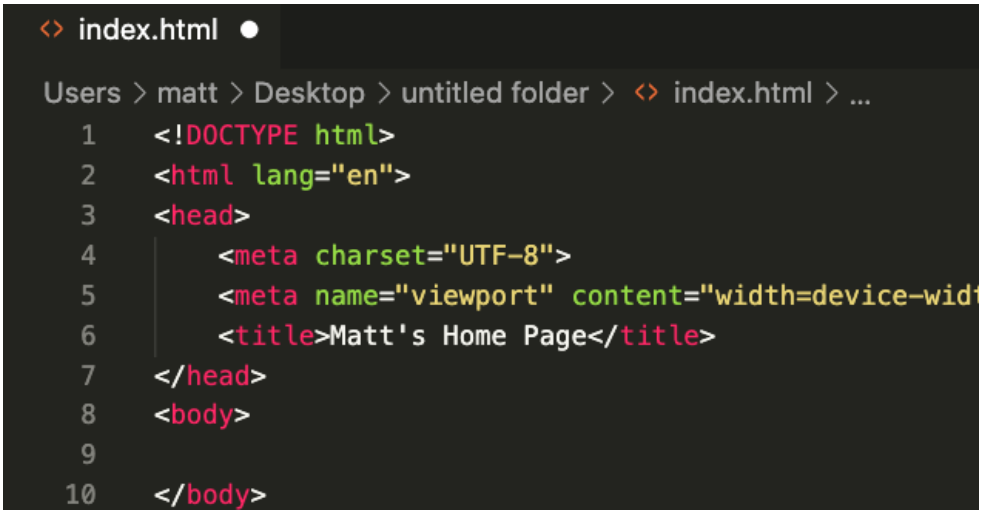


2. Select it and click enter. This will create a basic html 5 boiler plate document with a **doctype,** a **head** and a **body** for us with the basic shown below.

3. Save this as **index.html** into the “week1” folder in COMP 1589 on your G drive or One drive. This is going to be your home page (the first page that most people will see on your web site).

4. Change line 6 *Document* to *your name’s main page*. Eg 'Matt's main page'

4. Notice a dot is by the filename. Save the file (ctrl-s), what happens?



5. Move the cursor to line 9.
6. Now type in **`<h`** VS code is using code-hinting to give you some options. Select h1, pressing enter will select the highlighted elements and save you (a bit of typing)

```
7   </head>
8   <body>
9       <h
10          h1
11  </body  h2
12  </html  h3
            h4
            h5
            h6
            head
            header
            hgroup
            hr
            html
```

7. Now add **`>`** and code hinting should close the tag to give you **`<h1></h1>`**

8. The cursor should be flashing between the new tags, Type in your name

```
6       <title>Matt's main page</title>
7   </head>
8   <body>
9       <h1>Matt</h1>
10
11  </body>
12  </html>
```

9. Move the cursor to the end of line 9 and click 'enter'.

10. Now, similar to 9 type in **`<p>.`** Again, it should autocomplete.

11. Now type in *Hello World* within the paragraph element.

12. Use ctrl-s to save the file once more.

13. Open windows explorer and find index.html.

14. Right click on the file, and click on the **Open with** menu that pops up. Now choose Chrome or Edge to view your web page.

15. Keep the browser open on the right side of your screen. Now back in VS code put an anchor tag inside a paragraph element to create a web link to the Greenwich website. Again, you will notice that VS code uses code hinting to help/hinder you.

```
8   <body>
9       <h1>Matt</h1>
10      <p>Hello World!</p>
11      <p><a href="http://www.gre.ac.uk">University of greenwich</a></p>
12  </body>
13  </html>
```

16. Save your file once more (ctrl-s)

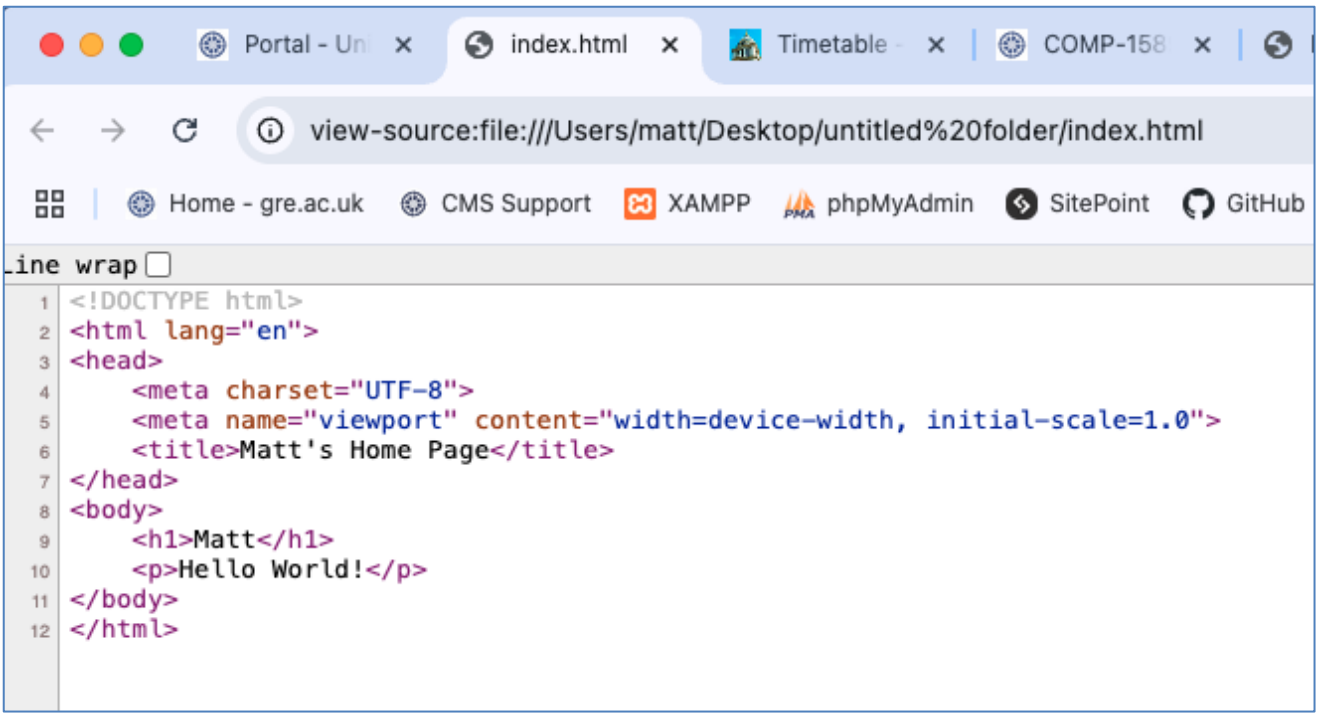
17. Return to your browser window and refresh your web page

18. Click on your link - does it work correctly? If not check for typing mistakes – try with and without target="_blank", what is the difference ?

```
10      <p>Hello World!</p>
11      <p><a href="http://www.gre.ac.uk" target="_blank">University of greenwich</a></p>
12  </body>
```

The power of VS code - VS code is a text editor. But gives you extra help as well.

Before we move on: Let's check out our source to understand the code a bit better, right click on the page and select "View Page Source". It should look something like the below.



Let's break this down line by line… Doctype

This is not actually a HTML tag, but rather information for the browser to understand what version of

```
<!DOCTYPE html>
```

HTML to treat this file as.

The above is for HTML5, generally this will be all you need for modern web development.

```
<html lang="en">
```

HTML

The HTML tag is the top level element of your HTML file. Within this we nest the <head> and <body> tags, which then hold most other tags. The lang attribute inside the opening tag sets the language for the page - this is beneficial for search engines and accessibility reasons, helping screen readers know how to properly pronounce the text.

Head

```
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Matt's Home Page</title>
</head>
```

This tag contains information that is processed by machines. Inside the <head> tags we nest our metadata which is data that describes the document to the machine.

Next we have one of our meta tags, we will look at these a little more over the coming weeks but these are special tags that allow us to describe how our page should be processed, displayed and interpreted by search engines (or similar tools).

Our first meta tag sets our page encoding, this determines what characters we can use. UTF-8 is the standard character encoding you should use in your web pages, this is a unicode standard that allows many languages and symbols. This will usually be the first <meta> tag shown in the <head> element

```
    <meta charset="UTF-8">
```

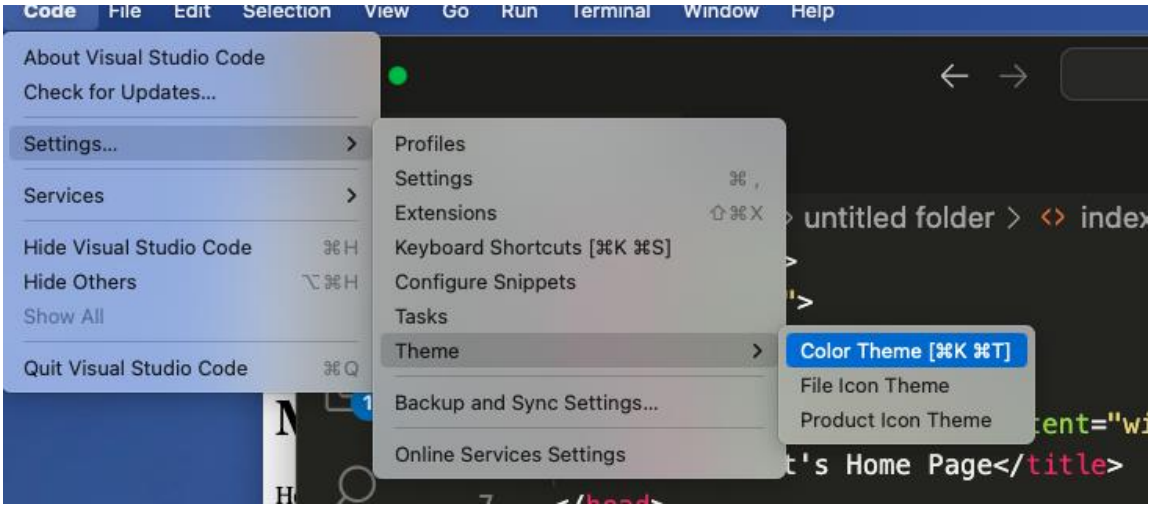
We then have our viewport meta tag, this determines how our webpage should respond across different screen sizes - this is known as responsive. The values used in this tag ensure our website width matches the viewport width on the device and our initial scale is 1 (100%).

```
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

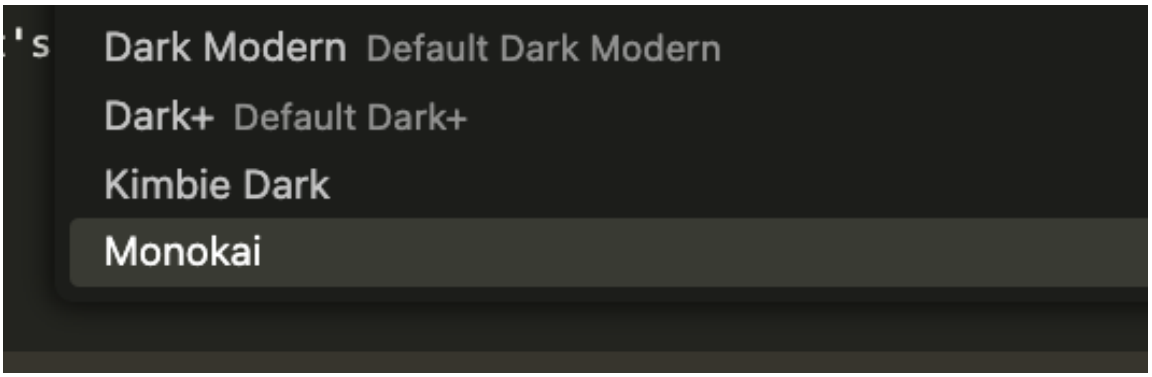
```
6    <title>Matt's Home Page</title>
```

Finally we have our <title> tag, this is the title for the web page and it is shown in the browser's title bar and sometimes used elsewhere such as in search engine results.

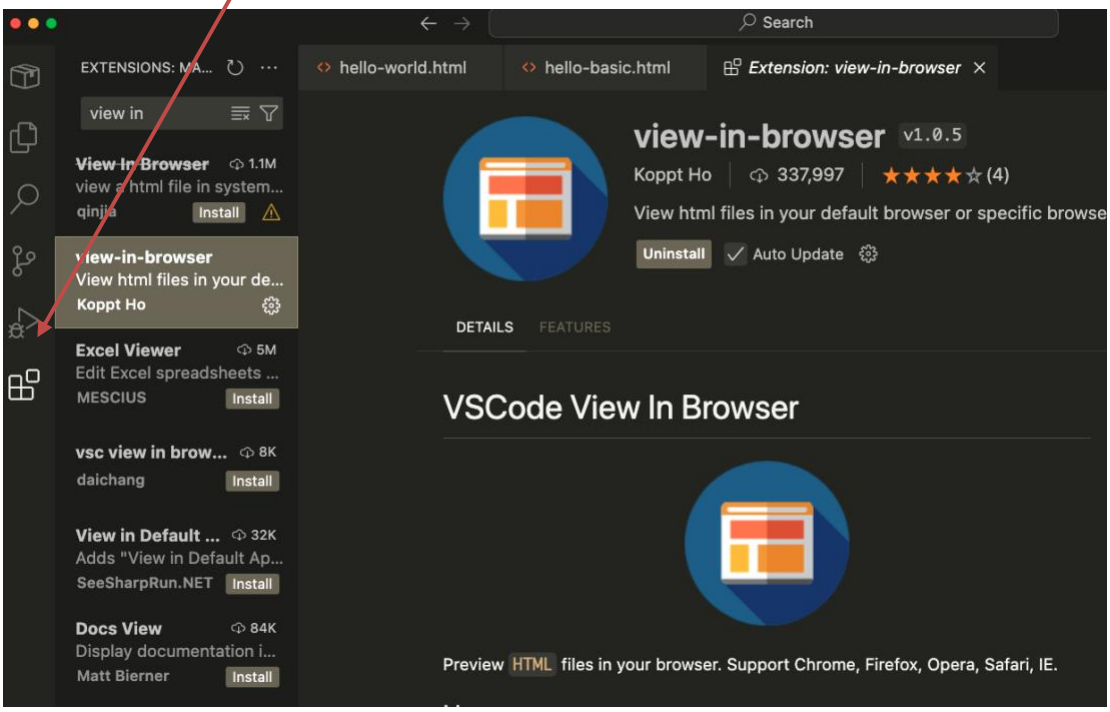
VS Code comes with many colour themes, go to >settings >Theme >Color Theme - there is a list to choose from.

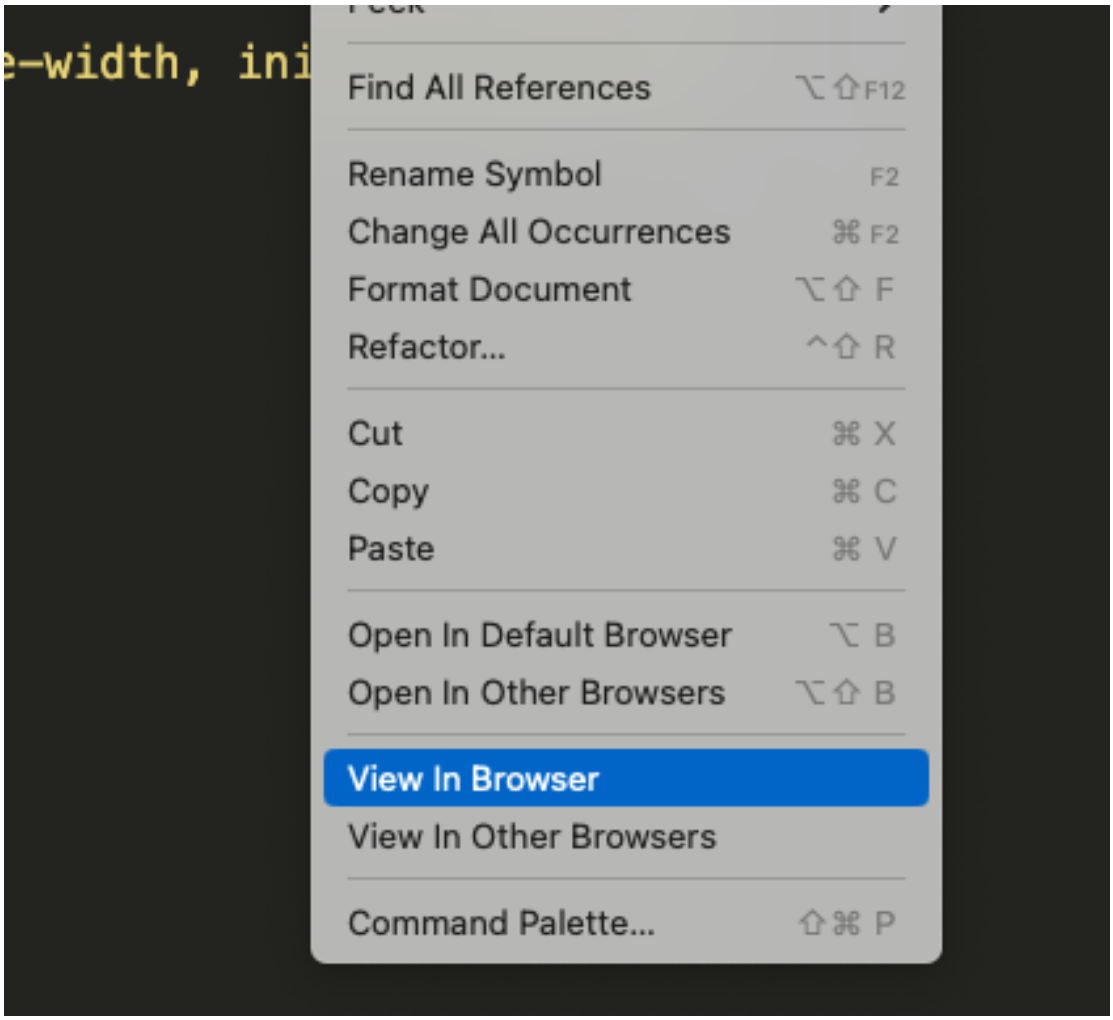


I use Monokai because I'm used to it now and I will use it in a lot of my slides, but it is up to you. Pick one and stick with it, as the colours help identify parts of the code.



If you search the extensions tab for view-in-browser, install it, then when you right click inside an open html document the view in browser menu item should be available





Explanation of some of the code

HTML files are text based files made up of elements / tags and attributes.
Some of the elements have to be there however you design the page
e.g. those in lines 1 to 8

Elements / tags

Look at line 2 and then the last line. This is the html element. It has an opening tag on line 2 and a closing tag as the last line. Every web page must have as its last tag the closing html tag </html>

Another rule of good practice is that tag names should be in lower case.

The html element contains two child elements:
<*head*> and <*body*>

The *head* element contains child elements that have general information about the web page (e.g. its title, author, any CSS styles used in formatting)

The *body* element is where you code most of your web page. It contains children that define the content of the web page (e.g. text images, links). **What goes in the body element is what appears in a browser.**

The body element is the parent of 3 elements in this web page: h1 and two p's

1. h1 marks its content (e.g. your name) as header text 1
2. p marks its content as paragraph text (e.g. *hello world* you just typed in)
3. the second paragraph is the parent of an anchor element. An anchor element forms the hyper text part (a *link*) of HTML.

Attributes in HTML

The opening tag of the anchor element is
`<a href="http://www.gre.ac.uk">`
In this example *href* is an attribute of the anchor tag.

There are some standard rules for attributes:

- Attributes only occur in opening tags
- Attribute names (eg href) must be in lower case
- Attribute values (e.g. **`"http://www.cms.gre.ac.uk/"`**) must be in quotes

The href attribute is the *hypertext reference* or URL (web address) for the link. So when you click on the link the browser will display that page.

Many tags have attributes, look at the *meta tag*, this has a number of attributes.
Some attributes are optional, some are compulsory

White space in HTML

Go back to VS code and your web page.
. Now add some extra spaces to the text *Hello World*. View in the browser
Make line 10 look like this

```
<p>Hell
o World!</p>
```

View in the browser
Try the same thing with the content of the anchor element. What if there is a space in the href attribute?

Why use VS Code?

Because we think it makes life easier for you. You can use notepad, and there are lots of other text editors out there such as Sublime Text.
HTML is stored as a text file you can use any text editor to create HTML (and CSS and JavaScript). This means that any piece of software that can read and write text can be used as an HTML editor (you could even use Word, but don't bother).
In CMS we use VS code because it gives a bit of extra help in writing HTML code, it is free, lots of extensions and works on PC's and Mac's (and Linux).

Recap:

The only stuff that is displayed in a basic HTML page must sit in the **body** element.
Pressing enter at the end of a line does not affect the display of text in a browser (except in a few special cases).

Colour coding

VS code (and many other packages) uses colour coding to help you identify tags and attribute

Look at the code in VS code.
What colours are :

- Tags
- Attributes names
- Values of attributes
- Text content of an element?

Adding more text to your web page

The paragraph element is the standard way of putting text into a web page. So replace the *Hello world* with a couple of sentences, explaining the purpose of this web site.

Save and preview in a browser.

Beneath this paragraph element type
<h2> Study</h2><h2> Interests</h2> <h2> Favourite web sites </h2><h2>work</h2>

Save and preview in browser.
Can you see how the header elements work? Header elements are part of the *structural (block type)* elements. They force a new line start in a browser display.
In VS code for ease of reading make each heading start a new line, with a few blanks line between.

Paragraphs and Breaks

Move the cursor in VS code to one of the blank lines below your *Study* heading.

Create a paragraph element. Within the paragraph tags type in the names of the courses you are studying this term, make each start a new line in VS code.

Save and preview in your browser.
The names of the courses should be displayed on the same line. To make them appear on separate lines yet still be part of the same paragraph you can use an extra tag. It's called the break tag <br />.
At the end of each course name add a break tag. Save and preview the effect in your browser.

Note: There isn't an opening and closing tag for the break element. It's been combined into one as it has no content. Its purpose is to force a new line start inside a block of text.
What is the difference between a paragraph in a browser and a break in a browser?

Anchor element

Under *favourite web sites* heading we want to create a set of links to other web sites.
Move the cursor to one of the blank lines under this heading.
Recall the anchor element is used to create a link to another page.

Type in **<a h**, press the down arrow and hit enter (selects href in the sub menu) type in (or paste) your favourite web address then close the bracket **>** and the closing tag will be entered for you

eg

```
<a href=" http://google.co.uk "></a>
```

Now enter the text that describes the link

e.g.

```
<a href="http://google.co.uk">UK google</a>
```
so it looks this in the browser

UK google

If you preview this in the browser it will work, however it is common practice to place the anchor inside a *structural* element (e.g. heading or paragraph element).

Create a paragraph element and then select and drag the closing tag

```
<a href="https://www.google.co.uk/">UK google</a>

<p></p><a href="https://www.google.co.uk/">UK google</a:

<p> <a href="https://www.google.co.uk/">UK google</a></p>
```

Add a few more links to your web page, making sure they are within paragraph elements.

Lists

In Word you can make bulleted and numbered lists. You can do something similar in HTML.

If you haven't done so already

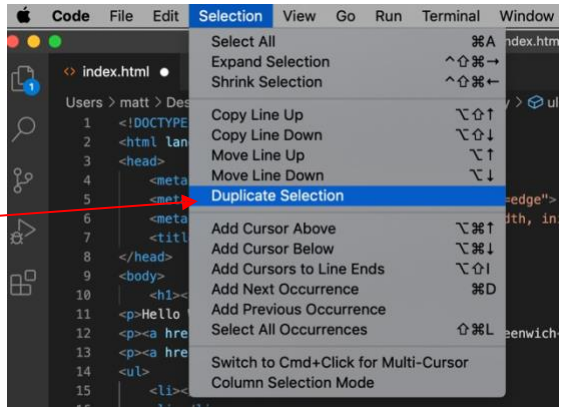
Enter **<h2> Interests</h2>**

Move the cursor to the line below

Read these instructions first and then follow them exactly

1. Type in `<ul>`
2. Press enter
3. Type in `<li>`
4. Press Ctrl-L (selects the line the cursor is in)
5. Press Ctrl-D (duplicates the line)
6. Press Ctrl-D again

   If Ctrl-D is not working select Duplicate Selection from the Selection menu



You will see this appear

```
<ul>
    <li> </li>
    <li> </li>
    <li> </li>
</ul>
```

This is using VS code to make you more efficient.

Add text between the **`li`** elements to actually create the list. As shown.

Note that you can add extra **`li`** elements, to add extra rows to your list.

```
<ul>
    <li>reading text books</li>
    <li>exams</li>
    <li>exercising</li>
</ul>
```

Save and preview in the browser.

Summary so far

You have been learning how to write valid, strict HTML in a text editor, and learnt some key terms. You have created the content and structure of a web page. The next step is to make the design and layout more exciting. For this you will need to learn about Cascading Style Sheets (CSS) next week.

Although there are many packages which make it easier to create web pages you will often have to return to looking at the HTML code to tweak it and sort out the errors they create. Even commercial packages now acknowledge this and they have sophisticated code editors as well as the WYSIWYG editor.

To do before the next tutorial

Install VS code on your home machine / laptop

Images to your home page

Find a small picture, of yourself, that you like. Save the file into your folder "week1" (the same folder as your web page).

In VS code, move the cursor to where you want the picture to appear (within the body element). Now type

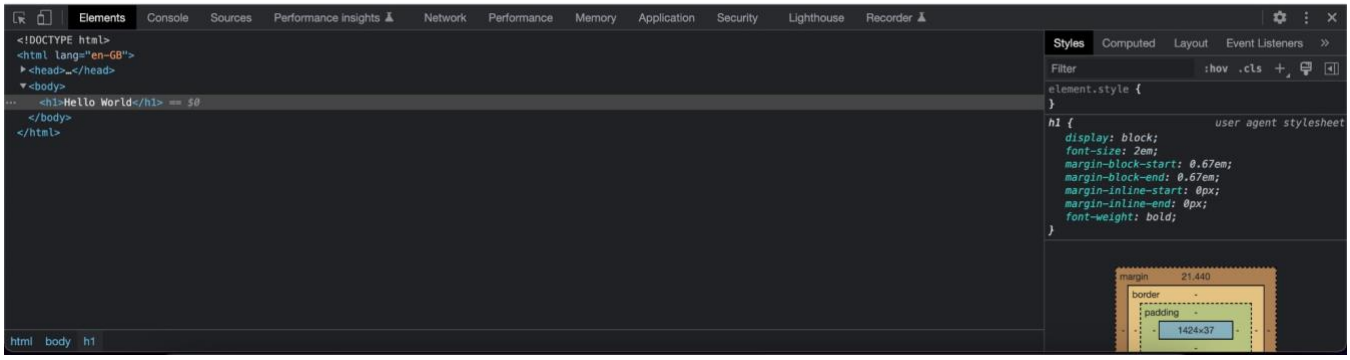
```
<p><img src="me.jpg" alt="" /></p>
```

Note that the alt attribute is mandatory and so you **MUST** put a brief description of the picture in the alt attribute value. `me.jpg` is the file name of the photo. Change this to match your image file name

Find another image and put the code for that inside the same paragraph tag. Save and view in browser. What happens as you resize your browser window.

You can finish up by exploring the browsers developer tools, these are a critical part of web development and for this part of the task I'd strongly recommend using Google Chrome. Within your browser, open your web page (using "File", Open; or simply drag and drop your HTML file into the browser). Now right click on the text and click "Inspect".

This will open up our developer tools, which should look something like:



We won't go into detail on these right now but please do have a play around in your own time to get a good understanding.

The main concepts to understand at this moment are the "elements" tab and the "styles" sidebar, these help you understand how your HTML code is viewed by the browser (this is not the same as "View Page Source" even though it looks it, this is what our code looks like after the browser has processed it).

Using the "elements" tab we can identify elements within the page and even change these to test out what things may look like, give it a try changing "Hello World" to "Hello Earth" in your H1 tag. Once you've done this, refresh the page and you'll notice your change is simply forgotten.

Within the "style" sidebar you can see the default styling the browser is applying, try clicking on the "body" element and you should see some default margin (space) and if you hover over that element it should highlight this in the browser and show that margin. We can also add our own custom styles in this sidebar, we will look at that more next week